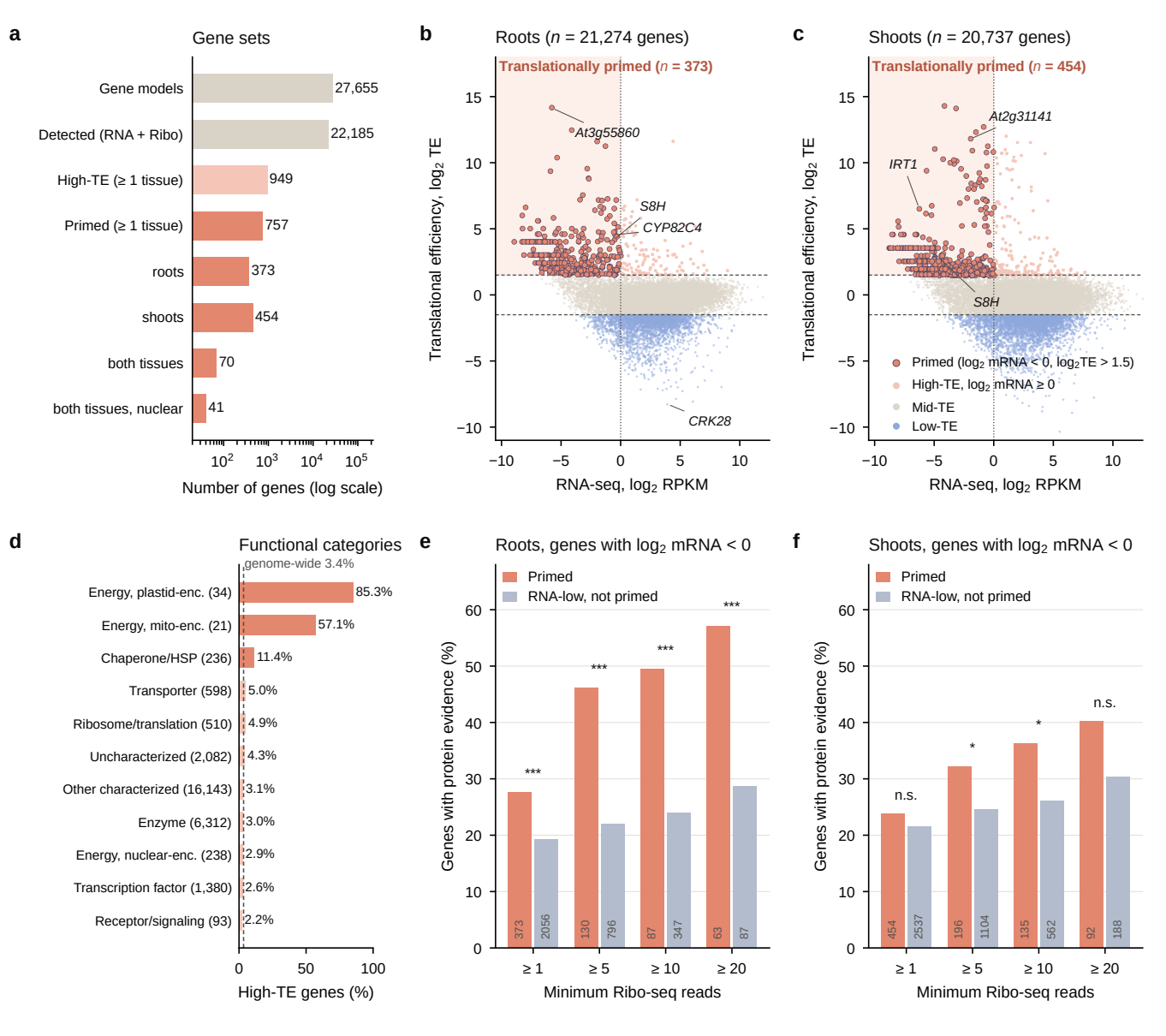

a
Gene sets
Gene models
27,655
Detected (RNA + Ribo)
22,185
High-TE (≥ 1 tissue)
949
Primed (≥ 1 tissue)
757
roots
373
shoots
454
both tissues
70
both tissues, nuclear
41
10^2 10^3 10^4 10^5
Number of genes (log scale)
b
Roots (n = 21,274 genes)
Translationally primed (n = 373)
At3g55860
S8H
CYP82C4
CRK28
Translational efficiency, log2 TE
RNA-seq, log2 RPKM
c
Shoots (n = 20,737 genes)
Translationally primed (n = 454)
At2g31141
IRT1
S8H
Primed (log2 mRNA < 0, log2TE > 1.5)
High-TE, log2 mRNA ≥ 0
Mid-TE
Low-TE
Translational efficiency, log2 TE
RNA-seq, log2 RPKM
d
Functional categories
genome-wide 3.4%
Energy, plastid-enc. (34)
85.3%
Energy, mito-enc. (21)
57.1%
Chaperone/HSP (236)
11.4%
Transporter (598)
5.0%
Ribosome/translation (510)
4.9%
Uncharacterized (2,082)
4.3%
Other characterized (16,143)
3.1%
Enzyme (6,312)
3.0%
Energy, nuclear-enc. (238)
2.9%
Transcription factor (1,380)
2.6%
Receptor/signaling (93)
2.2%
High-TE genes (%)
e
Roots, genes with log2 mRNA < 0
Primed
RNA-low, not primed
Genes with protein evidence (%)
373
2056
130
796
87
347
63
87
Minimum Ribo-seq reads
f
Shoots, genes with log2 mRNA < 0
Primed
RNA-low, not primed
Genes with protein evidence (%)
454
2537
196
1104
135
562
92
188
Minimum Ribo-seq reads